

Jayacharan R

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Education

Vellore Institute Of Technology, *B.Tech in Computer Science (Data Science) – Vellore* | **CGPA: 9.27** July 2023 – present
Achyuta Public School, *CBSE, Highschool in Bio-Mat – Dindigul, TN* | **AISSCE: 89.6%, AISSE: 98%** June 2017 – Apr 2023

Projects

WikiMind: End-to-End Advanced Agentic RAG & Knowledge Engine June 2026 – present

- Boosted Recall@K by **30%** across **6.8M** articles (**21GB+**) via LangGraph Hybrid RAG (Dense + BM25 + Rerank).
- Reduced staleness to **<5 min** & boosted query accuracy by **25%** using Redis Knowledge Graphs & spaCy.
- Elevated answer accuracy to **88%** at **<800ms** latency by tuning Self-RAG state machines & NeMo.

Clean-Backup: Intelligent Media Automation & Cloud Sync Engine Jan 2026 – present

- Accelerated deduplication by **10× times** (**1,000+** images/s) & cut storage by **81%** via Rust pHash (PyO3).
- Integrated Google Drive sync across **64** REST API endpoints via OAuth2 and a **3-stage** Docker build.
- Automated atomic rollback for **50+** batch operations across **14** sessions using journaled SQLite.

Experience

AI-ML Intern, *Tata Consultancy Services - TCS – Chennai, TN, India* June 2026 – July 2026

- Engineered high-accuracy RAG search (**<3.5s** latency, **390 tok/s**) with FastAPI, LangChain, ChromaDB & Groq.
- Secured pipeline against prompt injections with **<15ms** overhead by implementing Aegis LLM guardrails.
- Orchestrated real-time Langfuse telemetry across **100%** of traces to optimize performance & token costs.

SAP Analytics Cloud Intern, *Blueprint Technologies Pvt Ltd – Bengaluru, KA, India* May 2025 – July 2025

- Delivered **2** SAP Analytics Cloud dashboards (**20+** visuals) via EDA, cleaning, live Google Drive sync
- Strengthened insights via **3** predictive models (regression, classification, time-series) & dimension calcs
- Resolved lat-long ID conflicts, boosting geo-mapping accuracy across **210** countries in SAP Analytics Cloud

Awards

Winner - Roboverse Ideathon 2025, *IEEE Robotics and Automation Society, VIT, Vellore* Jan 2025

- Architected AI surveillance prototype using YOLO (Ultralytics) for real-time anomaly & movement detection
- Strengthened crime-prevention capability via automated alerts, behavioral analysis, and event logging

Skills

Programming & Frameworks: Python, Java, C, C++, Bash, YAML, HTML, CSS, JS, Pillow, FFmpeg, Tkinter

AI/ML Frameworks: PyTorch, LangChain, Langfuse, RAG, LLM Observability, LLM Guardrails, Quantization, Fine-tuning, HuggingFace Embeddings, ONNX Runtime, Qdrant (Vector DB), Redis, NetworkX (Knowledge Graphs), Reciprocal Rank Fusion (RRF), Cross-Encoder Reranking, NVIDIA NeMo Guardrails

Software: SAP Analytics Cloud, Git Version Control, AWS Architecture, Docker Deployment, Arch Linux

Language: English (Professional proficiency), Tamil (Native proficiency), Japanese & Hindi (Limited proficiency)

Certifications

AWS Certified Solutions Architect – Associate Dec 2025

Microsoft Certified: Azure AI Engineer Associate - AI 102 Nov 2025

Microsoft Certified: Azure Data Fundamentals - DP 900 Oct 2025